Full automorphisms of the dihedral subdivision family

Human Compression Project

Abstract

Suppressing vertices of degree two reduces the (S_4, D_{16}, D_8) incidence graph to a cubic graph. Tutte's bound on cubic arc stabilizers then leaves at most an index-two extension of the specified simple group. The action on the conjugacy class of a vertex stabilizer excludes that extension. This proves the four full-automorphism assertions for this family on p. 386 of the fixed $\mathrm{PSL}(2, q)$ source, conditional only on its separately verified subgroup configurations and the classical inputs stated below.

1 Suppression and the normalizer argument

Lemma 1. *Let X be a finite connected bipartite simple graph with degrees three and two. Suppose a group G preserves types, is edge transitive and acts two-transitively on the three neighbors of each vertex of degree three. If there are more than two vertices of degree three, suppressing all vertices of degree two produces a connected simple cubic graph Y. Moreover,* $\mathrm{Aut}(X) = \mathrm{Aut}(Y)$ *under their natural actions, and G is arc transitive on Y.*

Proof. Replace each path (u, e, v), with e of degree two, by an edge uv. There are no loops because u, v are distinct neighbors in the original simple graph. A priori parallel edges are possible. At any vertex u, two of its three incident edges in the resulting multigraph may have the same other endpoint. Having the same endpoint is a G_u-invariant equivalence relation on those three edges. The two-transitive action is primitive, so if such a pair exists all three edges at u have one common endpoint v. Cubicity at v then makes these two vertices a whole connected component. This contradicts connectedness and the hypothesis on the number of vertices. Thus Y is simple and cubic; connectedness is preserved by suppression.

Every automorphism of X preserves the two degrees and hence induces an automorphism of Y. Since there are no parallel edges, a vertex of degree two is uniquely determined by its two neighbors, so the induced map is injective. Conversely an automorphism of Y extends uniquely to its subdivision by its action on edges. This proves the asserted group identification. Edge transitivity in X gives transitivity on incidences of vertices and edges in Y, which is arc transitivity. □

Lemma 2. *Let a group G act faithfully and transitively on a finite set Ω, and let its point stabilizer M satisfy* $\mathrm{N}_G(M) = M$. *If $G \triangleleft H \leq \mathrm{Sym}(\Omega)$, conjugation embeds H into* $\mathrm{Aut}(G)$. *Its image preserves the G-conjugacy class of M.*

Proof. An element c centralizing G sends a chosen point x to a point gx fixed by $M = G_x$. The two point stabilizers have the same order, so $M = G_{gx} = gMg^{-1}$ and $g \in \mathrm{N}_G(M) = M$. Therefore $cx = x$. Commutation with the transitive group now implies that c fixes every point, so $c = 1$. The kernel of conjugation from H to $\mathrm{Aut}(G)$ is thus trivial. An element $h \in H$ takes G_x to G_{hx}, and all point stabilizers are G-conjugate. This proves the last assertion. □

2 The specified family

We use the following classical inputs as premises, not as new results. For a finite connected simple cubic arc-transitive graph, Tutte's theorem gives a vertex stabilizer of order at most 48 in its full automorphism group. Indeed that full group acts regularly on the s-arcs for some $s \leq 5$, and there are $3 \cdot 2^{s-1}$ such arcs from a fixed vertex. The two parts of this result are in Tutte's 1947 and 1959 papers [2,3].

For an odd prime p in the cases below, $G = \mathrm{PSL}(2,p)$ is nonabelian simple, $\mathrm{Aut}(G) = \mathrm{PGL}(2,p)$, and the two G-conjugacy classes of subgroups isomorphic to S_4 are interchanged by every outer automorphism. These are the classical subgroup and automorphism facts used by the source; the last fact is explicitly recalled on its p. 386. The subgroups S_4 are maximal, hence self-normalizing: their normalizers could otherwise only be G, contradicting simplicity.

Theorem 3. *Let $p \in \{17, 31, 79, 97\}$. Suppose $M, L \leq G = \mathrm{PSL}(2,p)$ give one of the connected locally two-transitive coset graphs with*

$$M \cong S_4, \qquad L \cong D_{16}, \qquad M \cap L \cong D_8.$$

Then the full automorphism group of its incidence graph is exactly the given group G.

Proof. The two degrees are three and two. The half with stabilizer M has $[G : M] = p(p^2 - 1)/48 > 2$ vertices. Lemma 1 gives a connected simple cubic graph Y with the same full automorphism group H and with G arc transitive. The action of G on Y is faithful, since it is its action on G/M and the core of the proper subgroup M in the simple group G is trivial.

By Tutte's bound and orbit–stabilizer,

$$[H : G] = \frac{|H_x|}{|G_x|} \leq \frac{48}{24} = 2.$$

Thus G is normal in H, either trivially or as a subgroup of index two. Lemma 2 embeds H into $\mathrm{Aut}(G)$, with image preserving the conjugacy class of M. Every outer automorphism interchanges the two S_4 classes, so this image contains no outer automorphism. It follows that $H = G$. The equality transfers back to the incidence graph by Lemma 1. □

The proof applies to every configuration with these hypotheses; it does not inspect a graph separately for each listed prime. More generally it applies to any prime for which the same classical premises and subgroup configuration hold. The count of configurations and their fusion under outer automorphisms are separate assertions, treated by the subgroup classification. The conclusion here supplies an upper bound on the full graph group; merely observing that an outer automorphism interchanges two configurations would not have supplied that bound.

The source's cubic (S_4, S_4, D_8) graphs require a separate argument. Their specified simple group has two vertex orbits, so Tutte's bound cannot be applied to their full group without first establishing vertex transitivity. We make no such inference in this note. No whole-source completion, Lean formalization, new conjecture count or measured compression factor is claimed by this component.

References

[1] F. Buekenhout, J. De Saedeleer and D. Leemans, *On the rank two geometries of the groups* $\mathrm{PSL}(2,q)$*: part II*, Ars Mathematica Contemporanea **6** (2013), 365–388, p. 386.

[2] W. T. Tutte, *A family of cubical graphs*, Proc. Cambridge Philos. Soc. **43** (1947), 459–474, doi:10.1017/S0305004100023720.

[3] W. T. Tutte, *On the symmetry of cubic graphs*, Canadian J. Math. **11** (1959), 621–624, doi:10.4153/CJM-1959-057-2.